

Converting Between Decimal Degrees and Degrees-Minutes-Seconds (DMS)

Decimal to DMS Conversion

Multiply the decimal part of the degree by 60 to get minutes.

Take the whole number part of those minutes as your final minutes.

Multiply the remaining decimal part of the minutes by 60 to get seconds.

Round the seconds to your desired decimal place.

Worked Example 1: Decimal Degrees to DMS

Convert 32.425° to DMS.

$$0.425 \times 60 = 25.5 \text{ minutes}$$

25 minutes and 0.5 minute remaining

$$0.5 \times 60 = 30 \text{ seconds}$$

Answer: $32^\circ 25' 30''$

DMS to Decimal Conversion

Divide the seconds value by 3600.

Divide the minutes value by 60.

Add the degrees, the divided minutes, and the divided seconds together.

Worked Example 2: DMS to Decimal Degrees

Convert $15^\circ 45' 18''$ to decimal degrees.

$$15 + 45/60 + 18/3600 = 15 + 0.75 + 0.005 = 15.755^\circ$$

Part A: Convert Decimal Degrees to DMS

1. 45.5°

2. 12.125°

3. 89.75°

4. 214.22°

5. 63.875°

6. 151.34°

7. 7.625°

8. 275.183°

Part B: Convert DMS to Decimal Degrees

9. $22^{\circ} 30' 0''$

10. $50^{\circ} 15' 36''$

11. $5^{\circ} 10' 12''$

12. $110^{\circ} 45' 0''$

13. $78^{\circ} 20' 24''$

14. $33^{\circ} 12' 18''$

15. $145^{\circ} 55' 30''$

16. $270^{\circ} 5' 45''$

Part C: Mixed Practice

17. Convert 18.425° to DMS.

18. Convert $95^\circ 48' 12''$ to decimal degrees.

19. Convert 72.35° to DMS.

20. Convert $12^\circ 6' 0''$ to decimal degrees.

Answer Key

1. $45^{\circ} 30' 0''$
2. $12^{\circ} 7' 30''$
3. $89^{\circ} 45' 0''$
4. $214^{\circ} 13' 12''$
5. $63^{\circ} 52' 30''$
6. $151^{\circ} 20' 24''$
7. $7^{\circ} 37' 30''$
8. $275^{\circ} 10' 59''$ (approx)
9. 22.5°
10. 50.26°
11. 5.17°
12. 110.75°
13. 78.34°
14. 33.205°
15. 145.925°
16. 270.096°
17. $18^{\circ} 25' 30''$
18. 95.803°
19. $72^{\circ} 21' 0''$
20. 12.1°